

A Geometric Theory of Contextual Semantic Capacity: Concentration of Measure and Contextuality in Large Language Models

Karl Svozil ^{1,*}

¹*Institute for Theoretical Physics, TU Wien, Wiedner Hauptstrasse 8-10/136, 1040 Vienna, Austria*

(Dated: May 6, 2026)

High-dimensional concentration geometry supplies the kinematic capacity for contextual semantic encoding in Large Language Models; hinge vectors and sense subspaces are one explicit geometric model of that capacity. Drawing on quantum contextuality, we analyze a polysemous token as a static hinge vector that gates the readout of meaning from a $(d - 1)$ -dimensional sense subspace orthogonal to it. Lévy’s lemma guarantees that exponentially many such subspaces can share a common hinge. We frame this as a geometric decomposition of contextual encoding, not a competing architecture, and propose probes for whether trained Transformers realize the structure.

Keywords: Large Language Models, Embeddings, Polysemy, Quantum Contextuality, Concentration of Measure, Lévy’s Lemma, Contextual Frames, Hinge Vectors, Superposition

I. INTRODUCTION

Large Language Models (LLMs) operate on dense vector representations of tokens in high-dimensional spaces. The success of these systems on linguistic tasks has been accompanied by a body of folk-theoretical claims about how meaning is encoded geometrically: directions correspond to features, distances to similarity, and arithmetic operations on vectors to semantic analogies [1]. Yet the mathematical underpinnings of these claims remain only partly articulated.

This paper addresses one specific problem: *polysemy*, the fact that a single lexical item (e.g., “bank”) can carry multiple, mutually exclusive meanings depending on context. In lexical semantics, polysemy is distinguished from homonymy by the relatedness of senses [2]; for our purposes, the key feature is that a single surface form must access distinct semantic content in distinct contexts. Modern Transformer-based LLMs handle this by producing *dynamic* contextual embeddings: the vector representation of a token is iteratively transformed by self-attention to absorb information from surrounding tokens [3]. This approach is empirically successful but conceptually opaque, since the same surface symbol is mapped to different vectors depending on its sentential neighborhood.

We propose a geometric interpretation inspired by quantum contextuality [4, 5]. On this reading, a polysemous token can be analyzed as a single fixed unit vector—a *hinge vector*—that is shared as a common member of multiple contextual frames. The hinge vector itself does not change; what changes is the contextual frame against which the surrounding linguistic state is read out. Crucially, the hinge vector plays an operational role within the analysis: it *gates* the readout by setting its overall magnitude, while the specific sense is decoded from the projection of the contextual hidden state onto the hinge-perpendicular subspace.

We do not claim that this static formalism is a competing *architecture* to dynamic Transformer embeddings. Rather, we claim that any successful contextual encoding—static or dynamic—must respect the geometric structure described here, and that dynamic deformation in trained Transformers can be analyzed as implicit realization of the same hinge-and-frame organization. Layer activations that move “bank” toward different regions of embedding space in financial versus river contexts can be read as bringing a particular sense subspace into alignment with downstream readout heads. The empirical question is then not which architecture is correct, but whether trained models in fact organize their representations in this way. Section V C sets out specific probes by which this can be tested.

The thesis of this paper is that *high-dimensional concentration-of-measure geometry provides the kinematic capacity required for contextual semantic encoding*. Contextuality supplies the combinatorial architecture: a single hinge vector may be the common element of multiple incompatible sense subspaces. Lévy’s lemma, applied to the linear overlap, supplies the geometric mechanism: in high dimensions, almost all candidate sense directions in the hinge-perpendicular subspace are nearly orthogonal to one another, with deviations bounded by $2\exp(-cd\varepsilon^2)$. Together, these imply that exponentially many approximately independent sense subspaces can share a common hinge vector. The Johnson–Lindenstrauss lemma [6], often invoked in similar contexts, arises from the same concentration phenomenon.

The analysis is conceptually adjacent to the *superposition hypothesis* in mechanistic interpretability [7], which holds that neural networks pack many features into nearly-orthogonal directions in a low-dimensional space, with the count bounded by exactly the kind of concentration estimate we use. The present formalism differs in identifying a specific structural pattern—hinge-and-frame rather than unstructured overcomplete dictionary—and in foregrounding the gating role of the hinge vector. In a sparse-coding picture, polysemous to-

* karl.svozil@tuwien.ac.at

kens correspond to special *junction* directions where multiple feature subspaces intersect.

The paper is organized as follows. Section II develops the geometric foundations: quasi-orthogonality, Lévy’s lemma on the linear overlap, the relation to Johnson–Lindenstrauss, and the definition of stable contextual frames. Section III introduces hinge vectors, the gated-subspace readout of context states, the bridge between concentration and contextuality, a proposition counting distinguishable sense subspaces sharing a hinge vector, and the relation to dynamic Transformer embeddings. Section IV discusses self-attention as approximate frame readout, formulates frame selection as a latent-variable model, and analyzes the role of stochastic decoding. Section V addresses the relation to the superposition hypothesis, limitations including the anisotropy threat, empirical predictions, and the scope of the quantum analogy.

Notation. Throughout, $\mathbb{R}^d$ denotes Euclidean d -space with the standard inner product $\mathbf{u}^\top \mathbf{v}$ and induced norm $\|\mathbf{v}\| = \sqrt{\mathbf{v}^\top \mathbf{v}}$; $S^{d-1} = \{\mathbf{v} \in \mathbb{R}^d : \|\mathbf{v}\| = 1\}$ is the unit sphere; $\mathbf{I}$ is the identity matrix; $(\cdot)^\top$ denotes transpose; $\|\cdot\|_{\text{op}}$ is the operator (spectral) norm; $A \succeq 0$ means A is positive semi-definite; $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation under the uniform (Haar) measure on S^{d-1} unless otherwise noted; $O(\cdot)$ and $\Omega(\cdot)$ are the standard Landau symbols. We write $\text{ReLU}(x) = \max(0, x)$ and use σ for the logistic sigmoid $\sigma(x) = (1 + e^{-x})^{-1}$. Universal positive constants are denoted c , c' , possibly differing between occurrences.

II. CONCENTRATION GEOMETRY AND FRAME CAPACITY

A. Quasi-Orthogonality

Let $\mathbf{v}_1, \dots, \mathbf{v}_N \in \mathbb{R}^d$ be unit vectors. The vectors are *strictly orthogonal* if $\mathbf{v}_i^\top \mathbf{v}_j = 0$ for all $i \neq j$. By elementary linear algebra, at most $N = d$ such vectors can coexist in $\mathbb{R}^d$.

This linear bound is restrictive. Many semantic features that one might wish to encode as directions need not be perfectly distinguishable; it suffices that they be *nearly* distinguishable. We therefore introduce ε -quasi-orthogonality: a set of unit vectors is ε -quasi-orthogonal if

$$|\mathbf{v}_i^\top \mathbf{v}_j| \leq \varepsilon, \quad i \neq j. \quad (1)$$

Since exact orthogonality is neither achievable nor necessary in realistic embedding spaces, quasi-orthogonality is the operationally meaningful notion throughout this paper.

B. Lévy’s Lemma on the Linear Overlap

The exponential capacity of high-dimensional spaces for quasi-orthogonal vectors is closely related to

concentration-of-measure phenomena. In high dimensions, most of the volume of the unit sphere S^{d-1} lies near the equator relative to any fixed direction. Consequently, two independently sampled random unit vectors are almost orthogonal with overwhelming probability.

This may be formalized using Lévy’s lemma [8, 9]. Let $f: S^{d-1} \rightarrow \mathbb{R}$ be an L -Lipschitz function, i.e., $|f(\mathbf{x}) - f(\mathbf{y})| \leq L\|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in S^{d-1}$. Then for any $\delta > 0$,

$$\mathbb{P}(|f(\mathbf{v}) - \mathbb{E}[f]| \geq \delta) \leq 2 \exp\left(-\frac{cd\delta^2}{L^2}\right), \quad (2)$$

where $c > 0$ is a universal constant and the probability is taken over the uniform measure on S^{d-1} .

To obtain quasi-orthogonality, we apply Lévy’s lemma to the *linear* overlap $f_{\mathbf{u}}(\mathbf{v}) = \mathbf{u}^\top \mathbf{v}$, which is 1-Lipschitz by the Cauchy–Schwarz inequality. By symmetry, $\mathbb{E}[f_{\mathbf{u}}] = 0$. Therefore, for any $\varepsilon > 0$,

$$\mathbb{P}(|\mathbf{u}^\top \mathbf{v}| \geq \varepsilon) \leq 2 \exp(-cd\varepsilon^2). \quad (3)$$

For the standard normalized surface measure on S^{d-1} , an explicit form of the constant gives $2 \exp(-(d-1)\varepsilon^2/2)$ [8]. Two independent random unit vectors are therefore ε -quasi-orthogonal with probability at least $1 - 2 \exp(-cd\varepsilon^2)$, and typical overlap amplitudes scale as $d^{-1/2}$.

The geometric significance is immediate: in sufficiently high dimension, almost all directions are nearly orthogonal to any given direction, and this quasi-orthogonality is a robust, generic property of high-dimensional spheres.

C. Relation to the Johnson–Lindenstrauss Lemma

The concentration phenomenon underlying Eq. (3) is closely related to the Johnson–Lindenstrauss (JL) lemma [6], which states that any set of n points in Euclidean space can be projected into a space of dimension $k = O(\varepsilon^{-2} \log n)$ while preserving all pairwise distances to within a factor of $(1 \pm \varepsilon)$.

Conceptually, both results arise from the same high-dimensional geometric fact: random projections and random directions are highly regular in large dimension. The JL lemma emphasizes the stability of finite metric structure under random projection, whereas Lévy’s lemma emphasizes the concentration of Lipschitz observables around their mean values on high-dimensional spheres. The two are mathematically intertwined—one can derive the JL lemma from concentration of measure on the sphere applied to projection norms [10]—but they highlight complementary aspects of the same phenomenon.

For the present purposes, Lévy’s lemma is the more natural tool, because the relevant issue is not dimensionality reduction of a finite dataset, but the typical quasi-orthogonality and abundance of semantic directions in high-dimensional embedding spaces.

D. Stable Contextual Frames

A contextual frame is not merely a collection of pairwise quasi-orthogonal vectors. Pairwise bounds such as $|\mathbf{e}_i^\top \mathbf{e}_j| \leq \varepsilon$ do not guarantee a well-conditioned frame when d is large, because the cumulative spectral effect of many small off-diagonal entries can be significant. By the Gershgorin circle theorem, pairwise quasi-orthogonality only implies $\|\mathbf{G} - \mathbf{I}\|_{\text{op}} \leq (d-1)\varepsilon$, which is uninformative for $\varepsilon = O(1)$ and $d \sim 10^3$. We therefore require a stronger condition.

Definition 1 (ρ -stable contextual frame). A contextual frame $\mathcal{F}^{\mathcal{C}} = \{\mathbf{e}_1^{\mathcal{C}}, \dots, \mathbf{e}_d^{\mathcal{C}}\}$ with frame matrix $\mathbf{F}^{\mathcal{C}} = [\mathbf{e}_1^{\mathcal{C}} \dots \mathbf{e}_d^{\mathcal{C}}]$ is called ρ -stable if its Gram matrix $\mathbf{G}^{\mathcal{C}} = (\mathbf{F}^{\mathcal{C}})^\top \mathbf{F}^{\mathcal{C}}$ satisfies

$$\|\mathbf{G}^{\mathcal{C}} - \mathbf{I}\|_{\text{op}} \leq \rho < 1. \quad (4)$$

Exact orthonormal bases correspond to $\rho = 0$.

Stability ensures that all singular values of $\mathbf{F}^{\mathcal{C}}$ lie between $\sqrt{1-\rho}$ and $\sqrt{1+\rho}$. Consequently, the dual frame $\tilde{\mathbf{F}}^{\mathcal{C}} = \mathbf{F}^{\mathcal{C}}(\mathbf{G}^{\mathcal{C}})^{-1}$ is well-defined and satisfies $\|\tilde{\mathbf{F}}^{\mathcal{C}}\|_{\text{op}} \leq (1-\rho)^{-1/2}$. The dual-frame readout is therefore numerically stable.

E. Exponential Packing

The concentration bound (3) yields an exponential packing result for individual vectors. Sample N unit vectors independently and uniformly from S^{d-1} . By a union bound over the $\binom{N}{2} < N^2/2$ pairs,

$$\mathbb{P}(\exists i \neq j: |\mathbf{v}_i^\top \mathbf{v}_j| \geq \varepsilon) < N^2 \exp(-cd\varepsilon^2). \quad (5)$$

This probability is less than one—and hence an ε -quasi-orthogonal family of size N exists—provided

$$N < \exp\left(\frac{c}{2}d\varepsilon^2\right). \quad (6)$$

For typical LLM embedding dimensions ($d \sim 10^3$ to 10^4) and modest tolerances ($\varepsilon \sim 0.1$), this bound permits enormous numbers of quasi-orthogonal directions. We refer to this exponentially large supply as the *quasi-dimensionality* of the embedding space.

We emphasize, however, that the packing bound controls only pairwise overlaps. As the Gershgorin remark above shows, pairwise control does not by itself yield operator-norm control over a frame matrix. The frame-level analogue requires a separate construction, which we present below in Proposition 1.

III. HINGE VECTORS AND SENSE SUBSPACES

A. The Hinge-Vector Formalism

We propose the following formalism. Each token t is assigned a fixed unit vector $\mathbf{v}(t) \in \mathbb{R}^d$, called its *hinge*

vector. Each context $\mathcal{C}$ in which t appears is represented by a ρ -stable contextual frame $\mathcal{F}^{\mathcal{C}} = \{\mathbf{e}_1^{\mathcal{C}}, \dots, \mathbf{e}_d^{\mathcal{C}}\}$ with frame matrix $\mathbf{F}^{\mathcal{C}}$, organized so that

$$\mathbf{e}_1^{\mathcal{C}} = \mathbf{v}(t). \quad (7)$$

The remaining frame vectors $\{\mathbf{e}_2^{\mathcal{C}}, \dots, \mathbf{e}_d^{\mathcal{C}}\}$ are unit vectors in $\mathbf{v}(t)^\perp \subset \mathbb{R}^d$ and span a $(d-1)$ -dimensional *sense subspace*

$$\mathcal{S}^{\mathcal{C}} = \text{span}\{\mathbf{e}_2^{\mathcal{C}}, \dots, \mathbf{e}_d^{\mathcal{C}}\} \subset \mathbf{v}(t)^\perp. \quad (8)$$

A token is *polysemous* if its hinge vector belongs in this way to two or more contextual frames, each carrying a distinct sense subspace. For example, $\mathbf{v}(\text{bank})$ might serve as the hinge of a finance frame, a river frame, and a tilt-of-aircraft frame, each with its own sense subspace inside $\mathbf{v}(\text{bank})^\perp$.

The gated-subspace readout of context. The meaning of a polysemous token is not contained in its hinge vector alone, because in every frame that contains it, the hinge has the same coordinates: $(1, 0, \dots, 0)$. Meaning is relational. It emerges from the interaction between the hinge vector and the surrounding linguistic context, mediated by the sense subspace.

Given an input context x , let $\mathbf{h}(x) \in \mathbb{R}^d$ denote the contextual state—in a static model, a vector derived from the surrounding tokens; in a trained model, the hidden state produced by preceding layers. We decompose $\mathbf{h}(x)$ along the hinge:

$$\mathbf{h}(x) = h_{\parallel}(x) \mathbf{v}(t) + \mathbf{h}_{\perp}(x), \quad h_{\parallel}(x) = \mathbf{v}(t)^\top \mathbf{h}(x), \quad (9)$$

with $\mathbf{h}_{\perp}(x) \in \mathbf{v}(t)^\perp$. The two components play distinct roles: $h_{\parallel}(x)$ measures how strongly the context *engages* the token at all, while $\mathbf{h}_{\perp}(x)$ carries the *sense-disambiguating* information.

The dual-frame readout in frame $\mathcal{C}$ is performed within the sense subspace. Let $\tilde{\mathbf{F}}_{\mathcal{S}}^{\mathcal{C}} = [\tilde{\mathbf{e}}_2^{\mathcal{C}} \dots \tilde{\mathbf{e}}_d^{\mathcal{C}}]$ be the frame matrix restricted to the sense subspace, and let $\tilde{\mathbf{F}}_{\mathcal{S}}^{\mathcal{C}}$ denote its dual. Then

$$\boldsymbol{\alpha}^{\mathcal{C}}(x) = (\tilde{\mathbf{F}}_{\mathcal{S}}^{\mathcal{C}})^\top \mathbf{h}_{\perp}(x) \in \mathbb{R}^{d-1} \quad (10)$$

gives the dual-frame coefficients of $\mathbf{h}_{\perp}(x)$ in $\mathcal{S}^{\mathcal{C}}$.

The *sense score* of frame $\mathcal{C}$ for context x is

$$s_{\mathcal{C}}(x) = h_{\parallel}(x)^2 \cdot r_{\mathcal{C}}(\mathbf{h}_{\perp}(x)), \quad (11)$$

where $r_{\mathcal{C}}: \mathbf{v}(t)^\perp \rightarrow \mathbb{R}_{\geq 0}$ is a within-subspace selector: it scores how well the sense subspace $\mathcal{S}^{\mathcal{C}}$ aligns with $\mathbf{h}_{\perp}(x)$. Three natural choices for $r_{\mathcal{C}}$ are

1. the squared norm of the dual-frame expansion, $r_{\mathcal{C}} = \|\boldsymbol{\alpha}^{\mathcal{C}}(x)\|^2$, when the frame is characterized purely by its orientation;
2. the squared overlap with a designated direction $\mathbf{w}^{\mathcal{C}} \in \mathcal{S}^{\mathcal{C}}$, $r_{\mathcal{C}} = (\mathbf{w}^{\mathcal{C}\top} \mathbf{h}_{\perp}(x))^2$, when a single sense-distinguishing axis is appropriate; or

3. a learned positive functional, e.g. a quadratic form $\mathbf{h}_\perp(x)^\top A^C \mathbf{h}_\perp(x)$ with $A^C \succeq 0$, when training infers the relevant within-subspace structure.

The selected frame is

$$\mathcal{C}^*(x) = \arg \max_{\mathcal{C}} s_{\mathcal{C}}(x). \quad (12)$$

The meaning of token t in context x is the relational triple

$$(\mathbf{v}(t), \mathcal{C}^*(x), \boldsymbol{\alpha}^{\mathcal{C}^*(x)}(x)). \quad (13)$$

On the form of the gating prefactor. The choice $h_\parallel(x)^2$ in Eq. (11) is a representative member of a family of admissible gating functions $g(h_\parallel): \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$, all sharing two essential properties: (i) $g(0) = 0$, so that hidden states orthogonal to the hinge produce no readout, and (ii) g is non-decreasing in $|h_\parallel|$, so that stronger engagement with the hinge amplifies the score. Natural alternatives include $|h_\parallel|$, $\text{ReLU}(h_\parallel)$ (sign-sensitive, firing only on positive overlap), and learned smooth gates $\sigma(\beta h_\parallel)$. Squaring is the simplest scalar that is smooth, non-negative, and zero only at $h_\parallel = 0$, which motivates it as a default; it does, however, discard the sign of $h_\parallel$, which may itself carry information (e.g., in antonymic contexts). In a learning setting, g would naturally be inferred from data alongside the frames.

The multiplicative form $g(h_\parallel) \cdot r_{\mathcal{C}}(\mathbf{h}_\perp)$ is also a choice, distinguishable from additive alternatives such as $|h_\parallel| + r_{\mathcal{C}}$. Multiplicative gating expresses the conjunction “the token is engaged *and* this sense is the one that fits,” and ensures that no sense scores positively when engagement is absent; additive forms do not have this property. We emphasize that the multiplicative form does not assume statistical independence between $h_\parallel$ and $\mathbf{h}_\perp$ in any context distribution—it specifies a functional combination, not a factorization of a joint density. The form of the gating function and the shape of $r_{\mathcal{C}}$ are independently choosable; the role of the formalism is to constrain their product to vanish when the hinge is not engaged.[11]

Why the hinge vector matters. In this formalism, the hinge $\mathbf{v}(t)$ plays a genuinely operational role: through the prefactor $g(h_\parallel(x))$, it gates the entire readout. When the contextual state is orthogonal to $\mathbf{v}(t)$, no frame associated with t scores positively, and the token is not engaged by the context. When $h_\parallel(x)$ is large, the readout is amplified and the within-subspace selector $r_{\mathcal{C}}$ decides which sense applies. Different hinges (for different tokens) induce different decompositions $\mathbf{h} = h_\parallel \mathbf{v} + \mathbf{h}_\perp$, and hence different sense subspaces in $\mathbf{v}(t)^\perp$. The geometry of the hinge is therefore not mere bookkeeping: it is what determines which subspace of context is exposed for sense disambiguation.

B. Concentration as the Geometric Substrate of Contextuality

Quantum contextuality concerns the coexistence of multiple incompatible measurement bases sharing common vectors [4, 5, 12]. In the present setting, the analogous structure is a family of semantic contexts represented by distinct contextual frames that partially overlap through a shared hinge.

The central geometric question is whether sufficiently many nearly independent sense subspaces can coexist in $\mathbf{v}(t)^\perp \cong \mathbb{R}^{d-1}$ without destructive interference. This question has a sharp answer:

1. *Contextuality* provides the combinatorial architecture—the logical possibility that a single hinge vector simultaneously belongs to multiple incompatible frames, each defining a distinct sense subspace.
2. *Concentration of measure* provides the geometric capacity—the guarantee that in high dimensions, exponentially many such subspaces can coexist with only weak mutual interference.

By Lévy’s lemma applied within $\mathbf{v}(t)^\perp$, once the hinge vector is fixed, the overwhelming majority of candidate sense directions remain quasi-orthogonal to one another. A single hinge may therefore simultaneously support a large number of approximately independent sense subspaces. Concentration of measure supplies the geometric substrate that makes contextual encoding scalable.

The role played by concentration here is analogous to the role played by large Hilbert-space dimension in quantum contextuality constructions: the combinatorial richness of intertwining contexts becomes increasingly abundant as dimension grows. The present proposal differs, however, in two important respects. First, exact orthogonality is replaced by quasi-orthogonality, which is both geometrically typical and operationally sufficient for semantic disambiguation. Second, quantum contextuality carries physical content beyond geometry: the Kochen–Specker theorem [5] proves that no noncontextual hidden-variable model can reproduce quantum predictions, regardless of the dimension. In our semantic setting, no analogous no-go theorem has been established. The contextuality we invoke is structural—it concerns the combinatorial pattern of frame-sharing—rather than foundational in the sense of constraining hidden-variable theories.

C. Many Sense Subspaces Sharing One Hinge Vector

The packing bound (6) counts individual quasi-orthogonal vectors. The central claim of this paper, however, concerns sense *subspaces*—structured $(d-1)$ -dimensional complements of a fixed hinge—that share a

common hinge vector. The following proposition shows that exponentially many such subspaces exist with controlled mutual coherence.

Proposition 1 (Sense-subspace capacity). *Fix a unit vector $\mathbf{v}_0 \in S^{d-1}$. For any fixed $\varepsilon > 0$ and sufficiently large d , there exist*

$$M \geq \frac{1}{d-1} \exp(c'\varepsilon^2(d-1)) \quad (14)$$

orthonormal contextual frames $\mathcal{C}_1, \dots, \mathcal{C}_M$ such that each frame contains $\mathbf{v}_0$ as its first vector and, for any two distinct frames $\mathcal{C}_a \neq \mathcal{C}_b$,

$$\left| (\mathbf{e}_i^{\mathcal{C}_a})^\top \mathbf{e}_j^{\mathcal{C}_b} \right| \leq \varepsilon \quad (15)$$

for all non-shared frame vectors $i, j \geq 2$, with a universal constant $c' > 0$.

Proof. The orthogonal complement $\mathbf{v}_0^\perp$ has dimension $d-1$. Sample M independent Haar-random orthonormal bases of $\mathbf{v}_0^\perp$ and prepend $\mathbf{v}_0$ to each, yielding M orthonormal frames of $\mathbb{R}^d$ with shared first vector $\mathbf{v}_0$.

For any fixed vector $\mathbf{e}_i^{\mathcal{C}_a} \in \mathbf{v}_0^\perp$ (with $i \geq 2$), the marginal distribution of any single basis vector $\mathbf{e}_j^{\mathcal{C}_b}$ ($j \geq 2$) drawn from an independent Haar-random orthonormal basis of $\mathbf{v}_0^\perp$ is uniform on the unit sphere of $\mathbf{v}_0^\perp$. By Lévy’s lemma applied within this $(d-1)$ -dimensional sphere,

$$\mathbb{P}\left(\left| (\mathbf{e}_i^{\mathcal{C}_a})^\top \mathbf{e}_j^{\mathcal{C}_b} \right| \geq \varepsilon\right) \leq 2 \exp(-c(d-1)\varepsilon^2). \quad (16)$$

Within-frame correlations do not enter because the bound concerns only *cross*-frame pairs. There are at most $M^2(d-1)^2$ such pairs. By the union bound, all cross-frame overlaps satisfy $\leq \varepsilon$ with positive probability provided

$$M^2(d-1)^2 \cdot 2 \exp(-c(d-1)\varepsilon^2) < 1. \quad (17)$$

This is satisfied for $M = \lfloor (d-1)^{-1} \exp(c'\varepsilon^2(d-1)) \rfloor$ with a suitable universal constant $c' > 0$. $\square$

For typical LLM dimensions ($d \sim 10^3$) and tolerances ($\varepsilon \sim 0.1$), Proposition 1 implies that the number of available sense subspaces sharing a single hinge vector is exponentially large, far exceeding the number of distinct senses that any natural-language token carries.

The proposition as stated produces frames with $\rho = 0$ (exact orthonormality). A relaxed version permitting ρ -stable but non-orthonormal frames could be obtained by sampling near-orthonormal frames—e.g., perturbations of Haar-random bases or outputs of a soft-stability regularizer—and applying the same union-bound argument together with a stability bound. We do not pursue this extension here; the orthonormal case suffices to establish the capacity claim.

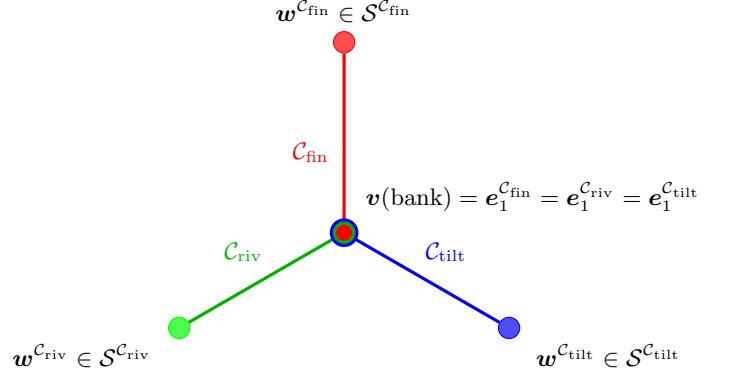


FIG. 1. Three intertwined contextual frames $\mathcal{C}_{\text{fin}}$, $\mathcal{C}_{\text{riv}}$, and $\mathcal{C}_{\text{tilt}}$ sharing the common hinge vector $\mathbf{v}(\text{bank})$ (financial institution, riverside, and aircraft-tilt senses). In each frame, the hinge is the first frame vector, $\mathbf{e}_1^{\mathcal{C}} = \mathbf{v}(\text{bank})$. Each frame carries a $(d-1)$ -dimensional sense subspace $\mathcal{S}^{\mathcal{C}} \subset \mathbf{v}(\text{bank})^\perp$. For visualization, a single distinguishing direction $\mathbf{w}^{\mathcal{C}} \in \mathcal{S}^{\mathcal{C}}$ within each sense subspace is shown. The semantic interpretation is not contained in $\mathbf{v}(\text{bank})$ alone, but in the gated readout of the surrounding context state through the hinge-perpendicular component (10).

D. Intertwining Contextual Frames

Two distinct contextual frames that share at least one common vector are called *intertwined*. Figure 1 illustrates a simple case: three contextual frames sharing a single hinge vector $\mathbf{v}(\text{bank})$, each carrying a distinct sense subspace within $\mathbf{v}(\text{bank})^\perp$. For visualization, a single distinguishing direction within each sense subspace is shown.

E. Relation to Dynamic Contextual Embeddings

Standard Transformer-based LLMs handle polysemy through dynamic embeddings. The token “bank” is initially mapped to a single static embedding $\mathbf{v}_0(\text{bank})$, but this vector is then transformed layer by layer through self-attention into a context-dependent vector $\mathbf{v}_L(\text{bank} \mid \text{context})$ that depends explicitly on the surrounding tokens.

The hinge-and-frame analysis is not in opposition to this picture. It offers a structural decomposition under which the dynamic trajectory $\mathbf{v}_0 \rightarrow \mathbf{v}_L$ can be read as implicitly performing frame selection: layer activations move the representation so as to bring a particular sense subspace into alignment with downstream readout heads. The static embedding plays the role of a candidate hinge; the layer-induced offset corresponds to the hinge-perpendicular component $\mathbf{h}_\perp(x)$; and the geometry of nearly-orthogonal sense subspaces explains why distinct senses can be cleanly separated by the network’s downstream computations. Table I summarizes how the two descriptions correspond.

TABLE I. Two descriptions of the same contextual phenomenon. The hinge-and-frame analysis is presented as a static decomposition of what dynamic Transformers do, not as a competing architecture.

	Dynamic description	Static decomposition
Token vector	Changes per context	Fixed (hinge) state
Context carrier	Layer activations	Hinge-perpendicular state
Polysemy mechanism	Vector deformation	Gated subspace readout
Geometric resource	Network depth	Quasi-dimensionality
Readout	Final-layer projection	Dual-frame coefficients

What would distinguish the two descriptions empirically? The hinge-and-frame analysis predicts that for a polysemous token, the projection of contextual hidden states onto the orthogonal complement of the static input embedding—i.e., the candidate hinge—should partition cleanly by sense. A purely dynamic account, by contrast, would not single out the static input embedding as a preferred direction; deformation could in principle proceed in any direction without privileging the original embedding. The prediction is therefore not about which architecture is correct, but about whether trained models exhibit hinge-anchored organization. Section V C formulates concrete probes for this question.

IV. ATTENTION, LATENT FRAMES, AND STOCHASTIC DECODING

A. Self-Attention as Approximate Frame Selection

The static contextual encoding of Section III is conceptually different from the dynamic embeddings of Transformers, but the two are not unrelated. Self-attention computes, for each token position, a convex combination of value vectors weighted by attention scores:

$$\mathbf{h}' = \sum_j \alpha_j \mathbf{v}_j, \quad \alpha_j = \frac{\exp(\mathbf{q}^\top \mathbf{k}_j / \sqrt{d})}{\sum_{j'} \exp(\mathbf{q}^\top \mathbf{k}_{j'} / \sqrt{d})}, \quad (18)$$

where $\mathbf{q}$, $\mathbf{k}_j$, and $\mathbf{v}_j$ are the query, key, and value vectors of standard self-attention, and $\mathbf{h}'$ is the attention output.

If the keys $\{\mathbf{k}_j\}$ are interpreted as frame vectors and the values as semantic content vectors, then the attention output is a weighted sum of frame elements. Any ρ -stable frame admits a dual-frame readout (Section II D); the tight-frame condition $\sum_j \mathbf{k}_j \mathbf{k}_j^\top = \lambda \mathbf{I}$ (with $\lambda > 0$) merely simplifies the dual to a scalar multiple of the primal. In that special case, the attention output (18) is particularly close in form to a dual-frame reconstruction with soft, data-dependent weights. More generally, the dual-frame readout is well-defined whenever the frame is well-conditioned, and attention can be viewed as a weighted aggregation that approximates this readout to varying degrees depending on how close the keys are to a tight frame.

Whether trained Transformers implicitly learn approximately tight frames—or, more generally, frames whose duals are well behaved—and whether the attention mechanism thereby approximates the dual-frame readout formalism, is an open empirical question. The static formalism provides a normative geometric target: if a network were to learn contextual frames, the dual-frame readout would be the correct decoding procedure.

B. Latent-Variable Frame Selection

Frame selection may be represented as a latent-variable decomposition:

$$p(t | x) = \sum_{\mathcal{C}} p(t | x, \mathcal{C}) p(\mathcal{C} | x). \quad (19)$$

Here $p(\mathcal{C} | x)$ is a learned distribution over contextual frames, and $p(t | x, \mathcal{C})$ is the token distribution obtained after reading the context state $\mathbf{h}(x)$ in frame $\mathcal{C}$.

Temperature may be introduced at the frame level,

$$p(\mathcal{C} | x) = \frac{\exp(s_{\mathcal{C}}(x)/T)}{\sum_{\mathcal{C}'} \exp(s_{\mathcal{C}'}(x)/T)}, \quad (20)$$

where $s_{\mathcal{C}}(x)$ is the gated sense score (11), or at the token level. Low temperature concentrates probability on the single frame whose sense subspace is most aligned with $\mathbf{h}_{\perp}(x)$ (and whose hinge has nontrivial overlap with $\mathbf{h}(x)$); high temperature spreads probability across multiple frames, permitting ambiguous or creative generation.

We note that the latent-variable formulation does not implicitly combine frames into a single composite frame matrix; mixing happens at the level of the probability distribution over $\mathcal{C}$, not at the level of vector concatenation. Cross-frame operator-norm control of the kind discussed in Section II D is therefore not required to make Eqs. (19) and (20) well-defined. It would become relevant only if a downstream computation explicitly read out from a stack of multiple frames simultaneously—a structure not assumed here.

C. Stochastic Decoding

LLMs generate text by sampling from a probability distribution over the vocabulary via the softmax map

$$p(t) = \frac{\exp(z_t/T)}{\sum_{t'} \exp(z_{t'}/T)}, \quad (21)$$

where $z_t = \mathbf{v}(t)^\top \mathbf{h}$ is the logit for token t given hidden state $\mathbf{h}$, and $T > 0$ is the temperature.

In the contextual framework, temperature controls a tradeoff between committed and ambiguous interpretations. In standard LLMs, the softmax (21) acts on token logits, not on a frame distribution. The two are

formally distinct: marginalizing the latent-variable expression (19) over frames yields a token distribution whose temperature behavior is not in general equivalent to applying a single temperature to the marginal logits. With this caveat, the qualitative picture is consistent: low effective temperature (whether applied at the frame or token level) concentrates probability on tokens consistent with the highest-scoring frame, yielding committed contextual readings, while high temperature spreads probability across frames and yields ambiguous output. Stochastic decoding is, in this sense, the mechanism through which genuine ambiguity can be expressed at the output stage, rather than being collapsed by deterministic argmax. Top- k and nucleus sampling truncate the distribution to a subset of likely tokens, and may be regarded as heuristics for restricting the search to high-scoring regions of frame-and-token space.

V. DISCUSSION

We have proposed that polysemy in LLMs may be understood geometrically as the participation of a single hinge vector in multiple contextual frames, with the hinge serving as a gate and the sense being decoded from a $(d-1)$ -dimensional sense subspace. Concentration of measure in high-dimensional spaces makes this scheme viable. The conceptual synthesis is:

- *Contextuality* supplies the combinatorial and logical structure—a hinge vector may be the common element of multiple incompatible sense subspaces.
- *Lévy’s lemma* supplies the geometric mechanism—in high dimensions, almost all candidate sense directions in $\mathbf{v}(t)^\perp$ are nearly orthogonal to one another.
- *Together*, they imply that exponentially many approximately independent sense subspaces can share a common hinge vector.

A. Relation to the Superposition Hypothesis

The proposal is closely related to the *superposition hypothesis* in mechanistic interpretability [7, 13], which holds that neural networks represent more features than they have neurons by packing those features into nearly-orthogonal directions. The capacity bound underlying superposition is precisely the concentration estimate (3) that we use here.

The two analyses share a geometric foundation but differ structurally. The superposition hypothesis treats the embedding space as an overcomplete dictionary of feature directions, with sparsity in the activation pattern enabling many features to coexist. The hinge-and-frame analysis identifies a specific organizational pattern within such a dictionary: features cluster around hinge vectors,

each hinge anchoring a structured family of sense subspaces. In a sparse-coding picture, polysemous tokens correspond to special *junction* directions where multiple feature subspaces intersect.

The relation between the two views is empirical. If trained models actually organize their feature directions into hinge-anchored families, then the structure of senses for a polysemous token should be readable from the geometry of its sense-conditioned activations, and dictionary-learning methods should recover such structure for known polysemous items. We leave this as an empirical question for mechanistic interpretability.

B. Limitations and Failure Modes

What the geometry provides is a quantitative explanation of why there is ample room for many contextual meanings to coexist without interference; it does not by itself establish that trained models exploit that room.

The framework can fail in several ways. First, training dynamics may constrain the effective embedding dimension or drive vectors into a low-dimensional submanifold. Empirical work has documented substantial *anisotropy* in trained contextual embeddings [14–16], indicating that representations occupy a narrow cone of the embedding space rather than exploiting the full quasi-dimensional capacity. This anisotropy directly reduces the effective d in our bounds.

The threat is quantitatively serious. Estimates of the intrinsic or participation dimension of contextual embeddings often yield values of order $10\text{--}10^2$ even when the nominal embedding dimension is $d \sim 10^3$. At $d_{\text{eff}} \sim 50$ and $\varepsilon = 0.1$, the exponent in our capacity bound is $c\varepsilon^2 d_{\text{eff}} \sim 0.5c$, a small constant rather than a large one—meaning that the exponential capacity claim collapses to “a few” in the worst case. For the framework to apply nontrivially, the geometry that matters is not the raw embedding space but its isotropized counterpart, obtained by whitening or related preprocessing techniques [15–17]. Empirically, such methods substantially restore high-dimensional geometry and improve downstream tasks; on the present account, this is precisely because they recover the room needed for contextual encoding. We regard the anisotropy threat as the single strongest caveat to the framework, and the question of which models, layers, and preprocessing regimes admit a meaningful d_{eff} as a priority for empirical work.

Second, the gated-subspace readout requires the frame matrix to be well-conditioned. Near-degeneracy ($\rho \rightarrow 1$) would amplify readout noise; in practice one would need $\rho \ll 1$ for numerical stability.

Third, the analysis identifies a specific organizational pattern (hinge-and-frame) within the broader superposition picture. If trained models do not in fact organize features in this way—if, for example, polysemous tokens are represented as superpositions without any preferred hinge direction—then the analysis would be descriptively

inaccurate even if its capacity bounds remain valid.

C. Empirical Predictions

The framework makes at least three testable predictions, all formulated against existing models and existing sense-tagged corpora (e.g., SemCor, WiC).

1. *Pairwise overlap distribution.* If trained embeddings exploit the available quasi-dimensionality, the distribution of pairwise inner products $\mathbf{v}_i^\top \mathbf{v}_j$ among token embeddings should be concentrated near zero with variance $\sim 1/d_{\text{eff}}$. Documented anisotropy implies that $d_{\text{eff}} \ll d$ in practice, providing a quantitative target for the gap.
2. *Sense subspaces in polysemous tokens.* For a known polysemous token, the projection of contextual hidden states onto the hinge-perpendicular subspace, stratified by sense, should cluster into approximately orthogonal subspaces of the embedding space. Concretely: take the static input embedding as a candidate hinge, project sense-stratified contextual representations onto its orthogonal complement, and test whether the resulting clusters lie in nearly-orthogonal linear subspaces.
3. *Hinge-gated readout.* Sense-discriminating linear probes trained on the hinge-perpendicular component $\mathbf{h}_\perp(x)$ should perform comparably to probes trained on the full $\mathbf{h}(x)$, while probes restricted to the parallel component $h_\parallel(x)\mathbf{v}(t)$ should perform near chance on sense classification. This would directly test the operational separation between gating (parallel) and disambiguation (perpendicular).
4. *Hinge-anchored organization (the critical test).* The analysis predicts that the static input embedding of a polysemous token is a privileged direction relative to its sense-conditioned contextual representations. Specifically: (a) the variance of sense-conditioned hidden states should be small in the direction of the static embedding and large in its orthogonal complement, and (b) clusters in the orthogonal complement should track sense distinctions more cleanly than clusters obtained by projecting onto arbitrary alternative directions. A null result here—if sense-conditioned hidden states do not single out the static input embedding as a special direction—would indicate that trained models perform contextual encoding without hinge-anchored organization, in which case the present analysis would remain valid as a description of *possible* structure but would fail as a description of *actual* structure in trained models.

D. Scope of the Quantum Analogy

The analogy to quantum contextuality is structural: both formalisms feature vectors that acquire interpretation through basis (frame) selection. However, the analogy has clear limits. Quantum contextuality carries physical content in the form of no-go theorems (Kochen–Specker, Bell) that constrain hidden-variable models; no such constraints have been established for semantic embeddings. The present framework uses the combinatorial architecture of contextuality—the pattern of frame-sharing—without inheriting its foundational implications. We adopt the contextuality vocabulary as a structural lens, not as a claim about hidden-variable incompleteness in semantic representations.

VI. CONCLUSION

We have proposed a geometric interpretation of contextual semantic capacity in LLMs. The thesis is that high-dimensional concentration-of-measure geometry provides the kinematic capacity required for contextual semantic encoding—a capacity that any contextual representation, static or dynamic, must respect. Quantum contextuality supplies the combinatorial structure of intertwined frames sharing common hinge vectors; Lévy’s lemma supplies the geometric mechanism that makes such structure scalable in high dimensions; and the Johnson–Lindenstrauss lemma appears as a sister manifestation of the same underlying concentration phenomenon.

Within the analysis, the hinge vector plays a genuinely operational role: it gates the readout of meaning by setting the magnitude of the sense score through its overlap with the contextual hidden state. The sense itself is decoded from the dual-frame expansion of the hinge-perpendicular component within a context-selected sense subspace. Different hinges induce different decompositions of the hidden state, so the geometry of $\mathbf{v}(t)$ directly determines which subspace of context is exposed for sense disambiguation.

The central quantitative content is the concentration estimate: for fixed $\mathbf{u} \in S^{d-1}$ and uniform $\mathbf{v} \in S^{d-1}$,

$$\mathbb{E}[\mathbf{u}^\top \mathbf{v}] = 0, \quad \mathbb{P}(|\mathbf{u}^\top \mathbf{v}| \geq \varepsilon) \leq 2e^{-c d \varepsilon^2}, \quad (22)$$

which implies that a single static hinge vector can anchor exponentially many distinct sense subspaces (Proposition 1). The gated-subspace readout (10)–(11) provides the mechanism for extracting context-dependent meaning from a fixed hinge.

We do not claim that this is how trained Transformers must work, nor that it is a competitive architectural alternative to dynamic embeddings. We claim instead that the hinge-and-frame structure is a natural geometric decomposition of the contextual encoding problem, that the structure is supported with room to spare by the geometry of high-dimensional spaces, and that whether

trained models realize the structure—rather than performing contextual encoding in some unstructured way—is an empirically tractable question. The connection to the superposition hypothesis suggests that mechanistic-interpretability methods are well-suited to test the analysis directly; the predictions in Section V C, particularly the hinge-anchored organization test, indicate where to look.

ACKNOWLEDGMENTS

This work was supported in part by the Austrian Science Fund (FWF) Grant DOI: 10.55776/PIN5424624. Portions of the manuscript were drafted with assistance from large language models; all ideas, arguments, and final formulations are the author’s responsibility.

-
- [1] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, Distributed representations of words and phrases and their compositionality, in *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 26 (2013) pp. 3111–3119.
 - [2] J. Pustejovsky, *The Generative Lexicon* (MIT Press, Cambridge, MA, 1995).
 - [3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, Attention is all you need, in *Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS’17* (Curran Associates Inc., Red Hook, NY, USA, 2017) pp. 6000–6010, arXiv:1706.03762.
 - [4] E. Specker, Die Logik nicht gleichzeitig entscheidbarer Aussagen, *Dialectica* **14**, 239 (1960), english translation at <https://arxiv.org/abs/1103.4537>, arXiv:1103.4537.
 - [5] S. Kochen and E. P. Specker, The problem of hidden variables in quantum mechanics, *Journal of Mathematics and Mechanics (now Indiana University Mathematics Journal)* **17**, 59 (1967).
 - [6] W. B. Johnson and J. Lindenstrauss, Extensions of Lipschitz mappings into a Hilbert space, in *Conference on Modern Analysis and Probability*, Contemporary Mathematics, Vol. 26 (American Mathematical Society, 1984) pp. 189–206.
 - [7] N. Elhage, T. Hume, C. Olsson, N. Schiefer, T. Henighan, S. Kravec, Z. Hatfield-Dodds, R. Lasenby, D. Drain, C. Chen, R. Grosse, S. McCandlish, J. Kaplan, D. Amodei, M. Wattenberg, and C. Olah, Toy models of superposition (2022), 2209.10652.
 - [8] M. Ledoux, *The Concentration of Measure Phenomenon* (American Mathematical Society, 2001).
 - [9] V. D. Milman and G. Schechtman, *Asymptotic Theory of Finite Dimensional Normed Spaces*, Lecture Notes in Mathematics, Vol. 1200 (Springer, Berlin, Heidelberg, 1986).
 - [10] S. Dasgupta and A. Gupta, An elementary proof of a theorem of Johnson and Lindenstrauss, *Random Structures & Algorithms* **22**, 60 (2003).
 - [11] For ease of exposition we treat the set of contextual frames as discrete, indexed by sense. In a learning setting, frames could equally be parameterized continuously—e.g., as elements of the Stiefel manifold $V_{d-1}(\mathbb{R}^d)$ —in which case Eqs. (11)–(12) become a continuous optimization rather than a discrete argmax. The discrete formulation is natural for sense disambiguation, where senses are categorical in lexicography, and we use it throughout.
 - [12] A. M. Gleason, Measures on the closed subspaces of a Hilbert space, *Journal of Mathematics and Mechanics (now Indiana University Mathematics Journal)* **6**, 885 (1957).
 - [13] T. Bricken, A. Templeton, J. Batson, B. Chen, A. Jermyn, T. Conerly, N. Turner, C. Anil, C. Denison, A. Askell, R. Lasenby, Y. Wu, S. Kravec, N. Schiefer, T. Maxwell, N. Joseph, Z. Hatfield-Dodds, A. Tamkin, K. Nguyen, B. McLean, J. E. Burke, T. Hume, S. Carter, T. Henighan, and C. Olah, Towards monosemanticity: Decomposing language models with dictionary learning (2023).
 - [14] K. Ethayarajh, How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings, in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (Association for Computational Linguistics, 2019) pp. 55–65.
 - [15] J. Gao, D. He, X. Tan, T. Qin, L. Wang, and T.-Y. Liu, Representation degeneration problem in training natural language generation models, in *International Conference on Learning Representations (ICLR)* (2019) 1907.12009.
 - [16] J. Mu and P. Viswanath, All-but-the-top: Simple and effective postprocessing for word representations, in *International Conference on Learning Representations (ICLR)* (2018) 1702.01417.
 - [17] J. Su, J. Cao, W. Liu, and Y. Ou, Whitening sentence representations for better semantics and faster retrieval (2021), 2103.15316.